AutoExam - Data Structures & Algorithms Oral Viva

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Sr.No	Question	Answer
1	What is the difference between a stack and a queue data structure?	A stack is a Last-In First-Out (LIFO) structure where insertions and deletions happen at the top. A queue is a First-In First-Out (FIFO) structure where insertion occurs at the rear and deletion occurs at the front.
2	Explain the average and worst-case time complexity of QuickSort.	QuickSort has an average-case time complexity of O(n log n). Its worst-case time complexity is O(n^2), which occurs when the selected pivot is repeatedly the smallest or largest element.
3	What are the key properties of a Binary Search Tree (BST)?	For every node X: all keys in the left subtree are strictly smaller than X.key, and all keys in the right subtree are strictly greater than X.key. In-order traversal of a BST yields sorted order.
4	How does a hash table resolve collisions using chaining?	Chaining stores multiple elements hashing to the same slot in a linked list attached to that hash bucket. Search traverses the linked list for the target key.
5	What is the fundamental difference between BFS and DFS in graph traversal?	Breadth-First Search (BFS) explores vertices level by level using a queue. Depth-First Search (DFS) explores as deep as possible along each branch before backtracking using a stack or recursion.